

Environmental Lawsuits against Homebuilders, Past and Present

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Abstract

How does the threat of a lawsuit affect a developer's decision to build new housing? I study residential permitting and construction patterns under two natural experiments involving State Environmental Policy Acts (SEPA). In the first, I show that passing a SEPA in the 1970s that ever takes a developer to an appeals court decreases new unit permitting rates by 11–45% through 1993, driven by fewer multifamily developments. In the second, a 2023 SEPA review “holiday” in Seattle provides encouraging but imprecise evidence that this effect is reversible. My results suggest that policymakers in SEPA states who want more housing built should protect most homebuilding projects from SEPA lawsuits.

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I Introduction

“The project they are attacking is probably quite costly to begin with; delay adds immeasurably to the cost. The point is to wear the defendants down. ... At some point, indeed, the defendants may count up the costs, look at the balance sheet, decide the fight is not worth it, and throw in the towel.”

Friedman (1989)

As the base of the second-largest real estate market in the world, the United States’ housing supply is enormous. It may also be misallocated: Hsieh and Moretti (2019) argue that local restrictions on land use in highly productive cities lowered aggregate US growth by 36% from 1964 to 2009. Existing studies of these local restrictions focus on variation in geographic and regulatory constraints on new housing.¹ In this paper, I investigate the role of an under-studied potential driver of regional differences in the housing supply: state and local variation in the risk of homebuilders being sued.

I use the passage of State Environmental Policy Acts (SEPAs) in the 1970s to investigate the effect an increase in litigation risk has on residential construction permit rates through 1993.² SEPAs require state governments to produce Environmental Impact Reports (EIRs) that detail the anticipated effects of government actions on the environment. EIRs must be shared with the public and can be challenged in court under circumstances that vary depending on the state. Crucially, in only half of all SEPA states has a residential development project ever been involved in a SEPA lawsuit that escalates to a court of appeals. I exploit this variation in a synthetic difference-in-differences framework to identify the effect of increased legal risk on the rate at which homebuilders acquire permits for new housing units, separate from the effect of just the SEPA itself. I find that SEPA passage decreases permitting rates for multifamily developments by an average of about one unit per 1,000 population per year in these high litigation risk states. SEPA passage has no effect on single family home permit rates in high risk states and does not have any effect at all in low litigation risk SEPA states.

I supplement these results by examining a 2023 natural experiment in Seattle that exempted it alone among cities in Washington from state EIR requirements. Anecdotal evidence suggests that developers in Seattle were not expecting this “review holiday”, allaying concerns about anticipation and selection effects. Over several specifications, I use other cities in Washington or other major US cities in a synthetic control framework to estimate the effect of Seattle’s review holiday on permit rates and its overall housing supply. Large positive results offer preliminary evidence that the chilling effect of SEPA lawsuit risk can be reversed

¹Examples include mountainous terrain (Saiz, 2010) and constructed indices of local land-use controls like minimum lot sizes (Glaeser & Ward, 2009). Glaeser and Gyourko (2018) provide a recent summary of many more studies in this vein.

²My historic specification ends at 1993 to avoid confounding with the significant expansion of credit access brought about by the Riegle-Neal Interstate Banking and Branching Efficiency Act. The effects of this law on the housing market are detailed by Favara and Imbs (2015) and de Chaisemartin and D’Haultfœuille (2026).

by targeted exemptions, but limited post-2023 data mean they lack precision.³

The first contribution of this study is to provide plausibly causal estimates of the effects of SEPA lawsuits to current policy debates. Historical accounts detail the scene in California after their state Supreme Court ruled in 1972 that EIR requirements applied to local governments:

“The state Supreme Court’s Sept. 21 decision requiring environmental impact reports on private construction left many labor leaders, builders, financiers, and public officials in shock and confusion. [...] Causing a good part of the confusion is the *lack of solid statistics* on a statewide basis on just how many building permits are being held up because of the ruling...”

“Confusion on Impact Reports Gets Worse”, Los Angeles Times, 22 October 1972; emphasis added.

50 years later, modern efforts to reduce the prevalence of SEPA lawsuits against homebuilders are currently spreading across SEPA states.⁴ Critics of reform efforts tend to share example cases when SEPA delay of construction proved valuable, especially in perceived wins for environmental causes. Proponents of reform efforts argue that they will spur housing development but can cite only descriptive or otherwise unidentified estimates to back their claims. By putting forth a “solid statistic” on the average impact of SEPA lawsuit risk on permitting rates, I contribute information that will be useful to participants in this increasingly relevant policy debate.

The second contribution of this study is to extend and deepen existing literature on the interplay between regulatory environments and housing supply in the United States. Mayer and Somerville (2000a) show that regulations that add delay or uncertainty for builders are the most likely to reduce development. This is the exact problem a builder would have with litigation: a lawsuit adds uncertainty to development both in the length of the proceedings (themselves a delay!) and in the chance of losing the case. Since builders are known to optimize for the regulatory environment in which they operate (Luger & Temkin, 2000), measures of changes in local regulations can enable causal analysis (Mayer & Somerville, 2000b). The Wharton Residential Land Use Regulatory Index (Gyourko et al., 2007, 2021) captures at least some of these effects with survey questions sent to municipalities about permit turnaround times and court involvement. My contribution to this effort is an objective measure of whether and the extent to which SEPA *lawsuits* affect homebuilders in a given state.

The third contribution of this study is to help answer a puzzle about litigation risk and capital investment in the field of law and economics. Arena and Julio (2015) use financial data on American firms to show that businesses reduce their capital expenditure when exposed to higher litigation risk as measured

³Recent advances in statistical inference for small-sample synthetic controls may increase the precision of these results. I plan to implement these new techniques in the very near term.

⁴Between 1 July 2025 and 26 May 2026, the following states updated their SEPA to limit the statewide impact of SEPA lawsuits on homebuilders: Minnesota, Washington, Connecticut, California, Massachusetts, and New York. See (future appendix table) for details.

by lawsuits against other firms in their industry. This finding suggests that firms hold on to capital when litigation risk increases. However, Chen et al. (2020) show that Nevada firms decreased capital expenditure when a new law substantially reduced their exposure to shareholder litigation risk. This finding suggests that firms hold on to capital when litigation risk *decreases*.⁵ Acquiring residential construction permits is itself a decision about capital expenditure: before applying for unit permits, developers must purchase and prepare a given piece of land. When SEPA laws pass and litigation risk increases, how do homebuilders react? My results suggest they hold on to their capital.

The full paper proceeds as follows. Section II describes the context that motivates the paper. The paper then continues in two large sections: Section III analyzes historical trends in residential permitting as SEPA laws spread across the United States in the 1970s, and Section IV analyzes an environmental review holiday that began in Seattle, Washington in 2023. Subsections in Sections III and IV establish relevant background details, discuss relevant data and summary statistics, explain identification strategies, and present main results and robustness checks pertaining to each larger section. Section V concludes the paper with a unifying discussion of the past and present results.

II Context

Residential development firms build and sell houses for profit. Figure 1 adapts figures from previous work outlining the decision to develop (Luger & Temkin, 2000; Mayer & Somerville, 2000a). They maximize expected profits and take housing market demand, input costs, and their available resources as given. They should also be expected to internalize the regulatory environment under which they plan to operate (Khan et al., 2024; Luger & Temkin, 2000). Highlighted by exclamation points in Figure 1 are steps at which developers interact with local government; that interaction is central to this study.

I focus my analysis of developers’ response to litigation risk on a class of laws known as State Environmental Policy Acts (SEPA laws). These laws were inspired by the National Environmental Policy Act of 1959 (NEPA), and they give the public the right to be informed of the potential environmental impacts of government projects. The mechanism legislators wrote into NEPA and the SEPA laws to share that information is usually known as an Environmental Impact Report (EIR).⁶ Government projects subject to EIR requirements under NEPA or a SEPA must first be assigned to a level of EIR, usually either a statement of no impact, a

⁵The authors’ finding holds most strongly for firms with a principal-agent problem between themselves as agents and their lenders as principals. Recently released data on bank real estate loans beginning in 1959 will allow me to explore this more deeply in planned work for this study.

⁶Exact names of the reporting tools vary, but they are fundamentally identical mechanisms designed to share information on predicted environmental impacts.

limited review, or a full EIR. This decision must be justified publicly, and the government then carries out whatever level of EIR is determined to be appropriate.

The two-stage EIR process of picking a level and carrying out review means *every* project covered by a SEPA is subject to at least some interaction with its EIR requirements. If a member of the public believes the government has not complied with EIR requirements (by assigning the wrong level of EIR or incorrectly completing the assigned level), they can assert that belief in court by bringing a SEPA lawsuit against the government. If the project involved in that case is, say, the decision to issue a permit to a residential development project, its developer becomes subject to the outcome of the lawsuit.

Though every SEPA is based in principle on the same federal law, their contents vary substantially. Both the kinds of projects that are subject to Environmental Impact Reporting (EIR) requirements under a given SEPA and who is allowed to bring a SEPA challenge in court vary by state according to both the actual text of a given law and that state’s courts’ interpretation of the law. Laws that open developers to litigation risk introduce uncertainty to the total costs and timeline associated with the decision to develop. After all, long lawsuits are expensive, and limits on litigation timelines are rare. Further information on the history and details of the exact laws will follow in each section below; the main point of interest is that exogenous changes to the kinds of projects that are subject to EIR requirements (by the introduction of a law or its later amendments) change the litigation risk under which a residential developer operates.

The variation in text and interpretation of existing SEPAs across states is wide, as will be discussed in further sections. This existing variation across the two criteria described above creates natural “blocks” of states that one can use to study the quantitative effects of a SEPA. Then, when a certain type of project is one day exempted from SEPA review requirements, that sudden change can also be exploited for productive analysis. Changing the expected cost of development by increasing or decreasing SEPA litigation risk in different places across time provides the variation that makes quantitative, causal evidence obtainable in this context: Mayer and Somerville (2000b) demonstrate that “new housing construction [is] a function of *changes* in house prices and costs rather than ... the levels of those variables” (emphasis added).

III The Past: The SEPA Seventies

III.1 Background

The National Environmental Policy Act (NEPA) went into effect January 1, 1970 in the midst of a wave of environmental activism spurred by the Santa Barbara oil spill, the Cuyahoga River fire, and other environ-

mental disasters. The law innovated environmental impact reviews (EIR) as a tool to encourage stewardship in federal decisionmaking, “require[ing] federal agencies to assess the environmental effects of their proposed actions” by preparing reports for projects with the potential to significantly affect the environment. The law also requires that these reports be shared with the public to solicit their input. NEPA was soon followed by the passage of SEPAs and other more limited EIR requirements in several states. Figure 2 categorizes states by the type of EIR requirements they passed in response to NEPA, following Ma et al. (2009) with a few adjustments.⁷ States with SEPAs apply their EIR requirements statewide, while those with Some EIR may have geographically or otherwise limited the extent to which EIR is required in their state. Though the SEPAs resemble NEPA in their structure and public review requirements, there are key differences across SEPAs in the extent to which they apply to private residential development.

The main difference I focus on is whether the text of a state’s SEPA is sufficiently broad to have motivated a productive appeals court case involving a private residential development project. As described by Hylton and Kim (2026), if the losing party in a lawsuit is sufficiently pessimistic about their chance of having the decision overturned, they will not appeal the decision to a higher court. Losers in SEPA suits that involve a residential development project are either governments issuing permits to developers with high sunk costs or opponents of that development that are of sufficient means and dedication to fight a court case. Strong feelings and incentives are clearly present on both sides of such a case. If a state has never had such a SEPA case make its way to an appeals court, then one of two conditions must be true: either 1) the SEPA clearly states that its requirements do not extend to private projects that need government permits and such a case therefore has never been brought to any court in that state; or 2) judges in that state have clearly signalled that decisions in any such SEPA cases have no chance of being overturned in a superior court. Under either condition, the effect is the same: private developers are under no SEPA litigation risk.

To that end, Table 1 lists every state with a SEPA, the section of that SEPA cited most frequently in court decisions reported in West’s National Reporter System (NRS), the number of court decisions in the NRS that cite that section of the SEPA, and (if applicable) the reference, year, and outcome of the earliest case in the NRS decided under that SEPA that concerned plans for residential development. Figure 3 shows the states from Figure 2 that passed SEPAs and the years in which they were passed, colored according to whether they have a case in the right panel of Table 1. SEPAs in states labeled “Appealers” have been demonstrated to be appeal-worthy in a way that SEPAs in “Never Appealed” states are not. By comparing

⁷First, while they treat New Jersey as having a SEPA due to a state executive order, I always treat New Jersey as having a limited EIR requirement. An executive order cannot give a plaintiff standing under which to bring a legal case against anyone but those carrying out the executive order, so finding case law for New Jersey’s “SEPA” is not possible, which warrants reclassification. Michigan, Texas, and Utah all also passed SEPA-like executive orders which have since been repealed (Pendall, 1998), but they are already classified as Tier 2 states due to their remaining limited EIR requirements.

estimates of the effect of passing a SEPA on homebuilding for each of these two groups, I separate the effect of having a SEPA from the effect of introducing SEPA litigation risk. The relevant research question is as follows: when states opened their developers to SEPA lawsuits, did their developers decide to build less housing?

III.2 Data & Summary Statistics

Building permits are public records in the United States. Since 1959, the Census Bureau has tracked building permits issued across the United States using the Building Permits Survey, a monthly survey conducted of jurisdictions within the United States that issue building permits. The historical data available to download from the Census Bureau track total state permit counts by year and extend from 1960 to present day. The sampling frame for the survey covers all “permit-issuing places” that do not double count. For example, if a city extends zoning permits and a county extends building permits for that city, only the county permitting office is surveyed. According to the Census Bureau, more than 99% of all privately-owned residential buildings are constructed in permit-issuing places.⁸

Each of the two blocks of SEPA case law described in Table 1 will be considered separately as treated groups. There remains the question of which states will eventually serve as the most appropriate comparison group to each set of SEPA states. Should states with limited EIR requirements like Nevada be included in the same pool of control states as Wyoming, where there are no EIR requirements at all? Table 2 displays summary statistics over relevant permit outcomes for the two blocks of SEPA states and the two blocks of non-SEPA states shown in Figure 2, broken down across pre-SEPA and post-SEPA time periods.⁹ It shows that, adjusted for population, states with SEPAs permitted housing at similar rates as their non-SEPA peer states. However, states whose SEPAs had residential appeals cases built multifamily housing at a higher rate (Mid/1K and Big/1K) than other states. Permitting rates for units in big multifamily developments decrease from pre- to post-SEPA in every EIR block, but single family home permitting rates (SFH/1K) only drop for the Appealers.

III.3 Identification Strategy

A standard difference-in-differences approach is the default starting point for evaluating the effects of changes in state policies over time. This approach relies on the parallel trends assumption that outcomes in treated

⁸A detailed explanation of the methodology behind the Building Permits Survey, as well as notes about imputation, reporting, and other data quirks associated with the BPS permitting data, can be found at U.S. Census Bureau (2026).

⁹Every SEPA was passed by 1975 with the exception of Georgia.

states 1) followed trends in untreated states before treatment; and 2) would have continued to follow the same trend as outcomes in untreated states had they been left untreated. This assumption is unlikely to hold in this context; there were likely many changes in housing trends across disparate regions of the United States, even if large ones like the spread of residential air conditioning had already promulgated by the 1970s (Lester, 2015). A large modern applied econometrics literature has also pointed out the bias and interpretability issues of the standard two-way fixed effects (TWFE) estimator commonly applied to these kinds of research designs when treatment rollout is staggered, as in this case (see Figure 3).

To illustrate the challenge of estimating causal effects in this context, consider the doubly robust “plug-in” estimator described in equation 4.2 of Callaway and Sant’Anna (2021).¹⁰ This estimator is designed to directly address issues arising from staggered treatment implementation, and it is available for easy use via Stata’s native `-xthdidiaregress-` package. If the parallel trends assumption holds unconditionally in this context (i.e. without control variables), the resulting estimates would be theoretically sound. Conducting a preliminary visual investigation of the validity of parallel trends in the pre-treatment period using this estimator is possible with the results in Figure 4. For the cohort of eight SEPA states with a history of residential lawsuits compared to all the non-SEPA states, the graphs suggest it is unlikely that parallel trends hold in the pre-treatment period, casting doubt on the validity of the unconditional parallel trends assumption needed for this estimator to be valid in this context.

Fortunately, there is an estimator that is robust to biases from heterogeneous treatment effects, staggered treatment implementation, and imperfect parallel trends: the synthetic difference-in-differences estimator (SDID) (Arkhangelsky et al., 2021). What follows is a description of the canonical and staggered treatment adoption SDID designs drawing heavily from sections 2.1–2.3 of Clarke et al. (2024), whose accompanying Stata package I use to conduct my analysis. SDID estimates the average effect $\hat{\tau}^{\text{SDID}}$ of a binary treatment W_{it} received by unit i at time t even in the face of potentially violated parallel pre-trends. The canonical SDID estimation solves the following minimization problem:

$$(\hat{\tau}^{\text{SDID}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau, \mu, \alpha, \beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \hat{\omega}_i^{\text{SDID}} \hat{\lambda}_t^{\text{SDID}} \right\} \quad (1)$$

where the right hand side is the ATT from a standard TWFE regression with unit fixed effects α_i and time period fixed effects β_t augmented with optimally selected unit weights $\hat{\omega}_i^{\text{SDID}}$ and time weights $\hat{\lambda}_t^{\text{SDID}}$. Details on the construction of $\hat{\omega}_i^{\text{SDID}}$ and $\hat{\lambda}_t^{\text{SDID}}$ are contained in both Arkhangelsky et al. (2021) and Clarke et al. (2024); they weight units and time periods such that the estimator matches treated and untreated units

¹⁰The estimating equation is not replicated here because it is not the primary focus of this section. Please refer to Callaway and Sant’Anna (2021) and the Stata help file for `-xthdidiaregress aipw-` for more information.

across time in a way that “synthesizes” parallel trends. In this context, μ is a constant, α_i and β_t represent state and year fixed effects, $\hat{\omega}_i^{\text{SDID}}$ weights states, $\hat{\lambda}_t^{\text{SDID}}$ weights years, and W_{it} is a binary indicator for having passed a SEPA.

As outcomes of interest Y_{it} , I use the number of housing units permitted per 1,000 people in a given state and year. I estimate the model separately for total units, single-family homes (SFH/1K), units in “mid-sized” buildings of 2-4 units (Mid/1K), and units in “big buildings” of 5 or more units (Big/1K). Accounting for population in the permit count outcomes mitigates the influence of baseline population differences and growth trends, which are among the largest contributors to demand in the housing market.¹¹ The sample opens at the beginning of statewide public permitting data in 1960 and ends in 1993. I end the sample just prior to the passing of the 1994 Interstate Banking and Branching Efficiency Act, which expanded bank branching rules, changed credit access, and had a significant effect on housing markets in many states (de Chaisemartin & D’Haultfoeulle, 2026; Favara & Imbs, 2015).

I estimate $\hat{\tau}^{\text{SDID}}$ separately for the two blocks of SEPA states: the eight states with a SEPA case that made it to an appeals court, and the seven states that have never had a residential case brought under their SEPAs. The choice of “donor pool” control states is not obvious *ex ante*. To this end, I alternate including in the control group 1) only those states with a limited-scope EIR requirement (Some EIR in Figure 2); 2) only those states with no EIR requirements; and 3) the full set of non-SEPA states (excluding New Mexico, as described above), giving a total of six estimates of $\hat{\tau}^{\text{SDID}}$ for each outcome in my main historical specification.

SDID is a particularly robust estimator. It inherits consistency from canonical difference-in-differences designs when the parallel trends assumption holds. In cases where that assumption fails for unadjusted data, it finds consistency by synthesizing unit and time weights to approximate parallel trends. This relies only on a donor convex hull assumption that it inherits from its roots in synthetic control methods – that the donor pool can be weighted to create a sufficiently well-fit counterfactual trend. Achieving consistency under just one of the two assumptions has led some to refer to SDID as “robust in a ‘double’ sense” (Nguyen, 2025), and this quality helps make estimated effects believable even when treatment is assigned as a function of unobserved unit characteristics. The tradeoff for a high likelihood of obtaining consistent estimates is reduced precision in finite samples; to account for the variance that arises from estimating coefficients *and* both sets of weights, SDID relies on a conservative standard error calculation that is described in Arkhangelsky et al. (2021).

¹¹This adjustment assumes stable preferences over housing services including household headship rates. More treatment of this adjustment will be available in future updates.

III.4 Results

Table 3 reports the results of synthetic difference-in-differences estimation of the historical effect of passing a SEPA across the several treatment and donor blocks described in Table 2, along with corresponding bootstrapped standard errors (10,000 replications). Each of the two panels treats one of the two blocks of SEPA states as treated and excludes the other. Estimates on different rows within a panel use a different set of non-SEPA states to construct the synthetic DID trend. Columns represent a different outcome variable, following the descriptions accompanying Table 2.

The most striking results are shown in columns 3 and 4 of the Appealed panel. They suggest that passing a SEPA that creates litigation risk causes substantial decreases in the rate at which developers secure permits for new housing in multifamily developments. The effect is consistent across donor pools and ranges between -0.90 and -1.38 units in Big developments per 1,000 population. Mid-sized developments also experience a significant decrease in permitting rates. These effects are completely absent for states whose SEPAs do not create litigation risk; if anything, the estimates in column 2 suggest developers in these states may actually build single family housing at a higher rate after a SEPA is passed. The pattern of coefficient estimates presented in Table 3 is consistent with litigation risk being the mechanism driving SEPA effects on permitting rates. It also suggests that litigation risk is a bigger issue for denser multifamily developments than it is for single family homes.

Examining the impact of the selected donor pool reveals an important trend. Estimates of the effects of SEPAs that use the No EIR donor pool are universally more negative than the estimates using the Some EIR and All non-SEPA donor pools. Table 2 already showed that the average No EIR state trends differently from averages in all other blocks by increasing total and single-family permits issued after 1975, so this effect ordering is not unexpected in those columns. No EIR states also comprise the only donor pool in which a lawsuit predicated on a residential developer’s public EIR disclosure is completely impossible, so these estimates may actually provide the most compelling counterfactual given the binary SEPA-on or SEPA-off assignment of treatment used here. The relative similarity of estimates across donor pools suggests this is not an issue of significant concern.

The economic implications of these estimates are large. What follows is a conservative but naïve counterfactual analysis. Suppose we take the average effect of passing a SEPA that creates litigation risk to be exactly one fewer unit permitted per 1,000 population, or a coefficient of -1. This estimate is conservative given that the sum total effect of SEPA passage on multifamily development rates (Mid/1K + Big/1K) in the Appealed panel is always greater than 1 in absolute value. Table 4 presents a naïve extrapolation of

this estimate, obtained by taking observed total unit permit rates, subtracting the coefficient estimate of -1, multiplying the resulting rate by population in a given year, and taking the average across the years after a SEPA passed in a given state. The ratio of observed average permits per year post-SEPA and the extrapolated counterfactual permits per year constitutes the main result of the paper: states with SEPA that impose litigation risk on developers permitted between 68.8–89.9% of the units they would have permitted otherwise. Said another way, the reciprocal ratio suggests that states with these SEPA would have permitted an average of 11.2–45.3% more housing units had their SEPA not introduced litigation risk to residential developers. This counterfactual exercise is a naïve extrapolation of a reduced form result, but it nevertheless demonstrates that the effects of SEPA litigation risk on historical residential development patterns was likely substantial.

IV The Present: A Review Holiday in Seattle

IV.1 Background

On 23 July 2023, Washington’s SEPA was updated by the state legislature to read as follows:

Until September 30, 2025, all project actions that propose to develop one or more residential housing or middle housing units within a city west of the crest of the Cascade mountains with a population of 700,000 or more are categorically exempt from the requirements of this chapter.

Wash. Rev. Code Ann. § 41.21C.229(4)

Seattle is the only city in Washington that meets these criteria. This single-city carveout – referred to in one source as an “environmental review holiday” (Trumm, 2026) – was particularly lenient, fully exempting all residential development within the city boundaries. By exempting Seattle development from SEPA requirements, the state legislature also exempted Seattle developers from SEPA lawsuits.

The Seattle environmental review holiday therefore affords a uniquely powerful opportunity to cleanly compare the effects of changes in SEPA litigation risk. Comparing a change in litigation risk across cities within a single state is likely to be as close to a unified regulatory environment as is possible for policy changes within the United States. There is also some evidence that construction patterns were not changing in anticipation of the holiday: projects that were under SEPA review at the time were allowed to withdraw the SEPA component of their applications (SDCI Community Engagement, 2023), suggesting that at least some developers were not anticipating this holiday when making the decision to begin the development process. The question for the present is therefore the inverse of the question for the past: when Seattle shielded its developers from liability for SEPA lawsuits, did they build more housing?

IV.2 Data & Summary Statistics

Washington state law assigns the responsibility of determining the populations of cities and towns within the state every year on April 1 to the state Office of Financial Management (Washington State Office of Financial Management, 2026). These population estimates are accompanied by estimates of the housing stock obtained using housing construction, completion, and demolition data reported by cities and counties to the state office.¹² Using estimates that incorporate completion and demolition data allows me to estimate the effect of SEPA changes on the housing stock as a whole, instead of being limited to just permitting as in the historical data. However, before 2020, official city-level estimates are not broken down by the type of housing unit, so the Seattle analysis is limited only to total housing unit counts. Despite this limitation, the extra information from construction and demolition data allow for richer examination of the effects of environmental litigation risk on the overall housing stock than would be possible with just the historical permitting data.

Comparing the housing stock during Seattle’s environmental review holiday to that of other incorporated municipalities within similar locales provides about as clean a test of the impacts of removing SEPA litigation risk as is possible in the United States, as it avoids many of the pronounced regional differences in land use regulations that are omnipresent across the many states of the nation. Nevertheless, the question of what cities constitute an appropriate comparison group is not easy to answer. Would the most appropriate comparison group be regional cities in the three counties that make up the Seattle-Tacoma-Bellevue metropolitan area? Perhaps the restrictive forces of building in sensitive coastal environments is most at play, so we should consider the set of cities in the fifteen counties in western Washington that border either the Puget Sound or the Pacific Ocean? Perhaps, owing to the fact that judicial interpretation of SEPAs plays a large role in the regulatory environment faced by builders, we should consider the six counties that make up Division I of the Washington Court of Appeals, where a cities’ SEPA cases would meet the same panel of judges as a Seattle SEPA case would?

Table 5 displays summary statistics of relevant outcomes for Seattle and the non-Seattle cities in three potential donor pools: the Seattle metro area, Washington counties that border the Pacific Ocean or the Puget Sound, and the counties in Division I of the Washington Court of Appeals. The data in this table make it clear that Seattle is the largest city in this region by a good margin, but the averages displayed overstate this comparison. Included with equal weight in the reported averages are very small towns with only a few thousand residents each; many more populous cities like Tacoma and Bellevue build at rates

¹²The OFM city population and housing stock estimates are available through the State of Washington Open Data site at data.wa.gov as the (Housing Units/Population) by State, County and City, 1990 to Present datasets. Accessible as of 27 July 2026, with data also available in the replication package for this paper.

similar to Seattle. The mix of geographically similar smaller towns with higher per 1,000 housing measures and other relatively large cities with lower per 1,000 statistics lends itself well to the identification strategy described in the next section.

IV.3 Identification Strategy

The classic use case for a synthetic control estimator is to estimate the effects of a treatment on a single treated unit among many, such as one region of Spain (Abadie & Gardeazabal, 2003), one state in the United States (Abadie et al., 2010), or one city in Washington state. As summarized by Abadie (2020), the synthetic control method is useful because, when comparing a single unit affected by treatment to potentially many others, “a combination of unaffected units often provides a more appropriate comparison than any single unaffected unit alone.”

For a set of cities C in western Washington state within a given time period t (here, a year), we observe an outcome of interest $Y_{c,t}$ (here, total housing units). Call $Y_{c,t}^N$ the total housing units in city c in year t when the treatment is Not active, and $Y_{c,t}^T$ the potential outcome when Treatment is active. The objective is to reliably estimate the following period-specific effect:

$$\tau_{c,t} = Y_{c,t}^T - Y_{c,t}^N \quad (2)$$

For every city but Seattle, $Y_{c,t}^T$ is assumed to equal $Y_{c,t}^N$, so that $\tau_{c,t} = 0$.¹³ For Seattle, $Y_{Seattle,t}^N$ is not observed in any time periods after treatment begins, so estimating $\hat{Y}_{Seattle,t}^N$ represents the key challenge addressed by a synthetic control.

A synthetic control Seattle is a weighted average of the other included cities’ total per capita housing counts. For a vector of non-Seattle city weights $\mathbf{W} = (w_1, w_2, \dots, w_{C-1})$ that are non-negative, sparse, and sum to one, the synthetic control estimators of $Y_{Seattle,t}^N$ and $\tau_{Seattle,t}$ are:

$$\hat{Y}_{Seattle,t}^N = \sum_{c \neq Seattle} w_c Y_{c,t} \quad (3)$$

and:

$$\hat{\tau}_{Seattle,t} = Y_{Seattle,t}^T - \hat{Y}_{Seattle,t}^N \quad (4)$$

The key problem then is how to calculate $\mathbf{W}$, the vector of city weights that build the synthetic

¹³This assumes that the Seattle review holiday had no effect on construction patterns in surrounding cities.

Seattle. Several methods exist for fitting these weights, including minimizing mean squared prediction error in the pre-treatment periods (as in Abadie and Gardeazabal (2003) and Abadie et al. (2010)) and ridge penalty selection (as implemented in Clarke et al. (2024)). Regardless of the fitting method, synthetic control methods rely on a convex hull assumption to identify causal treatment effects. The assumption is that a sufficiently well-fitting control “unit” can be constructed as a convex combination of units in the “donor pool” – the set of untreated units included as potential contributors to the synthetic control. This assumption does not have a formal test, but the traditional plots and figures associated with the exposition of a synthetic control serve as a valuable check. The Seattle outcomes I study are not extreme values in my dataset, so they plausibly fall within the convex hull of other Washington cities.

It is also helpful for the credibility of a synthetic control estimate to use in the donor pool those units that would already be considered similar to the treated unit. As categorized in Table 5, I report the results of estimation using three potential donor pools (in order from fewest to most counties covered): first, the set of cities in the three counties that make up the Seattle-Tacoma-Bellevue metropolitan statistical area; second, the set of cities in the six counties in Division I of the Washington Court of Appeals; and third, the set of cities in the fifteen counties that border either the Puget Sound or the Pacific Ocean. The list of counties covered by each donor pool is reported in the notes under Table 5.

Synthetic difference-in-differences is not only useful for improving difference-in-differences estimation with synthetic control tools. It can also improve synthetic control estimation with difference-in-differences tools. Though it may be the case that a given donor pool cannot convincingly synthesize a control for Seattle by synthetic control estimation, SDID may still be able to synthesize a control with parallel trends to Seattle. To check the robustness of each synthetic control estimate, I compare it to the corresponding SDID estimate. As the outcome of interest $Y_{c,t}$, I use the estimated total number of housing units per 1,000 people in a given city and year. Accounting for population in the outcome again mitigates the influence of baseline population differences and growth trends across cities. I choose to begin the time period of interest for this specification in 2010. This time period is selected as a natural normalization point for the housing market following the Great Recession.

IV.4 Results

Figure 5 displays the construction of synthetic Seattle using each of the three potential donor pools. It lists weights assigned to cities in each specification, with unlisted cities receiving zero weight. Each specification displays the constructed counterfactual unit and its pre-holiday fit to real Seattle as measured by mean

squared prediction error. Figure 6 displays those counterfactuals next to a corresponding SDID counterfactual trend constructed using the same donor pool. Finally, Table 6 lists the associated holiday effect coefficient estimates for synthetic control and SDID specifications.

Visual inspection of the synthetic controls in Figures 5 and 6 suggest that Seattle experienced a marked increase in its housing supply rate relative to its constructed counterfactuals. The coefficient estimates are quite large, ranging between 5-11 times larger in absolute value than the conservative 1 unit per 1,000 population estimate from the historical SEPA effect estimate. Alas, a sample of just three post-holiday treatment periods does not lend itself to much statistical power, and the synthetic control estimates are particularly imprecise.¹⁴ Nevertheless, the graphical evidence and large positive estimates are useful in at least one sense: they do not suggest that the SEPA holiday made fewer housing units available.

V Discussion & Conclusion

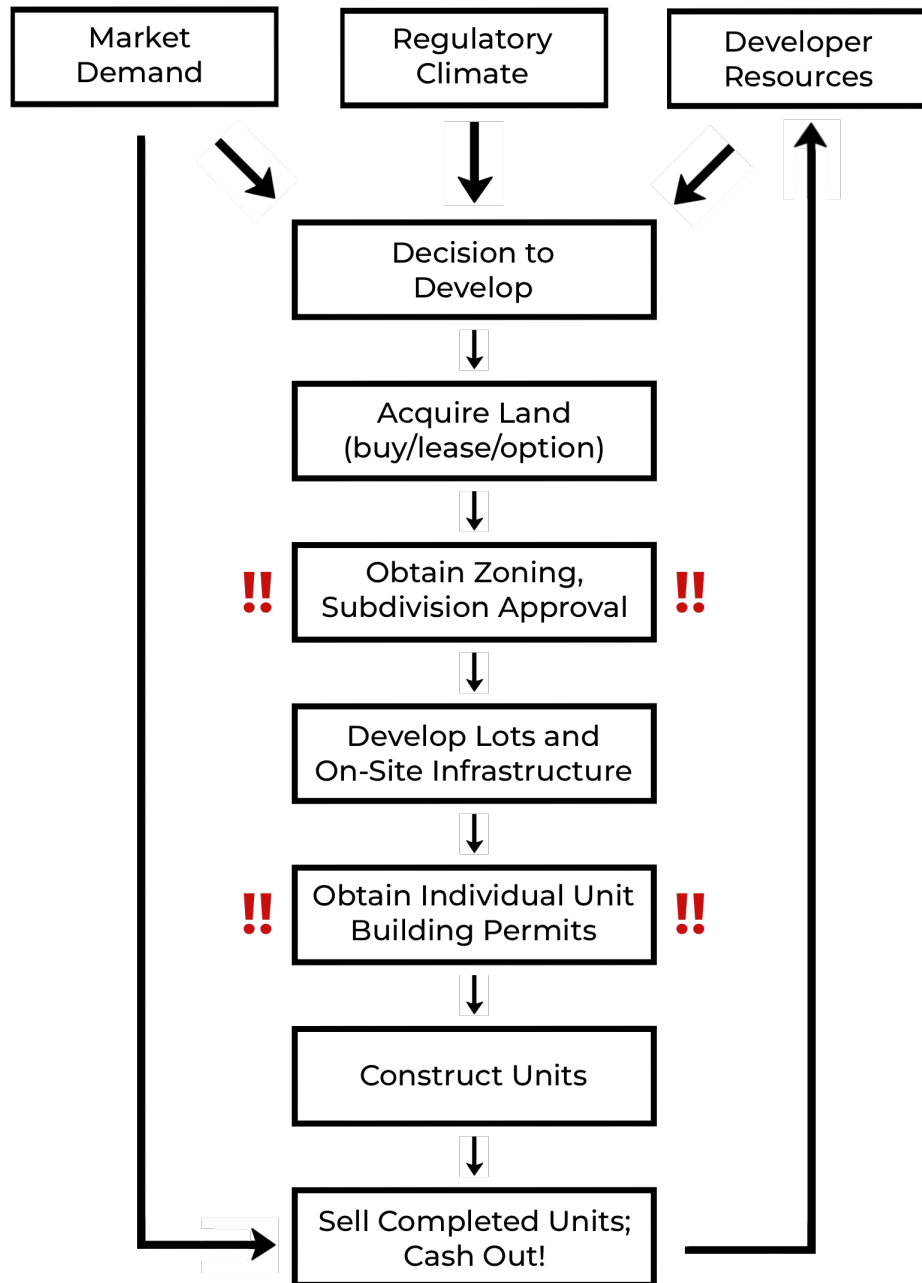
Taken together, the results from both the initial passage of the SEPAs in the 1970s and the 2023 Seattle EIR holiday tell a coherent story. The historical context suggests that SEPA litigation risk appears to have a strong chilling effect on residential development, especially for multifamily housing. A conservative yet naïve extrapolation of the historical results suggests that SEPA litigation risk reduced total per-capita permitting rates by 11–45%, depending on the state. These results are robust to the set of states included as donors to the synthetic trend used to identify coefficient estimates. The modern context provides noisy but encouraging evidence that these effects can be reversed by simply exempting residential development from EIR requirements.

These results make significant contributions to several fields of research. They give policymakers a “solid statistic” that will allow them to more fully investigate the costs and benefits when debating amendments to EIR requirements for residential development. They also show that homebuilders respond to judicial constraints, expanding on our existing understanding of how the housing supply responds to regulatory and geographic constraints. Finally, they show that homebuilders obtain fewer permits when litigation risk increases and more permits when it decreases, consistent with the hypothesis that the relationship between litigation risk and capital expenditure is negative for homebuilders. The results presented here paint a consistent picture: the delay and uncertainty associated with SEPA lawsuits chill homebuilding, but removing that risk may warm the chill. Future research into the effects of SEPAs on development at large will hopefully deepen this preliminary understanding over time.

¹⁴New methods for inference in small- t synthetic control settings are in the near-term plans for improvement of this paper. Plans to address potentially significant population movement (the denominator of the outcome) are also soon to be implemented.

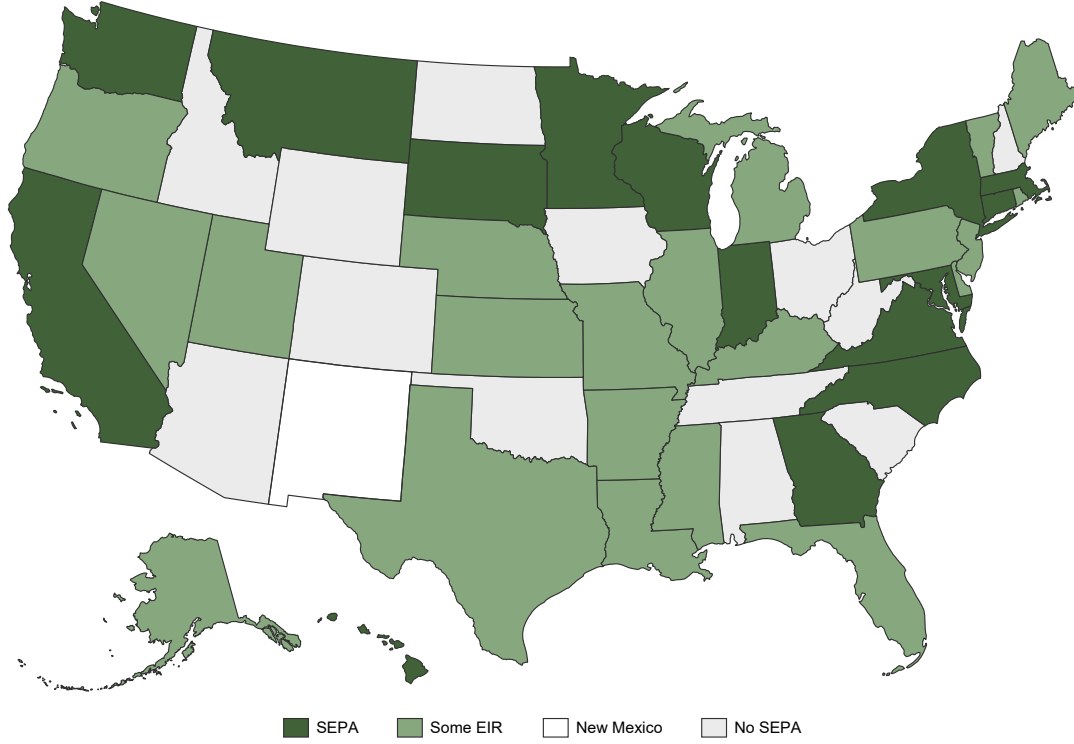
Tables & Figures

Figure 1: The Developer's Decision Calculus



Notes. This figure adapts and combines Figure 1 from Mayer and Somerville (2000a) and Figure 2 from Luger and Temkin (2000). Exclamation points denote stages of the process in which legal challenges are most likely to arise.

Figure 2: SEPA and EIR Requirement States, via Ma et al. (2009)



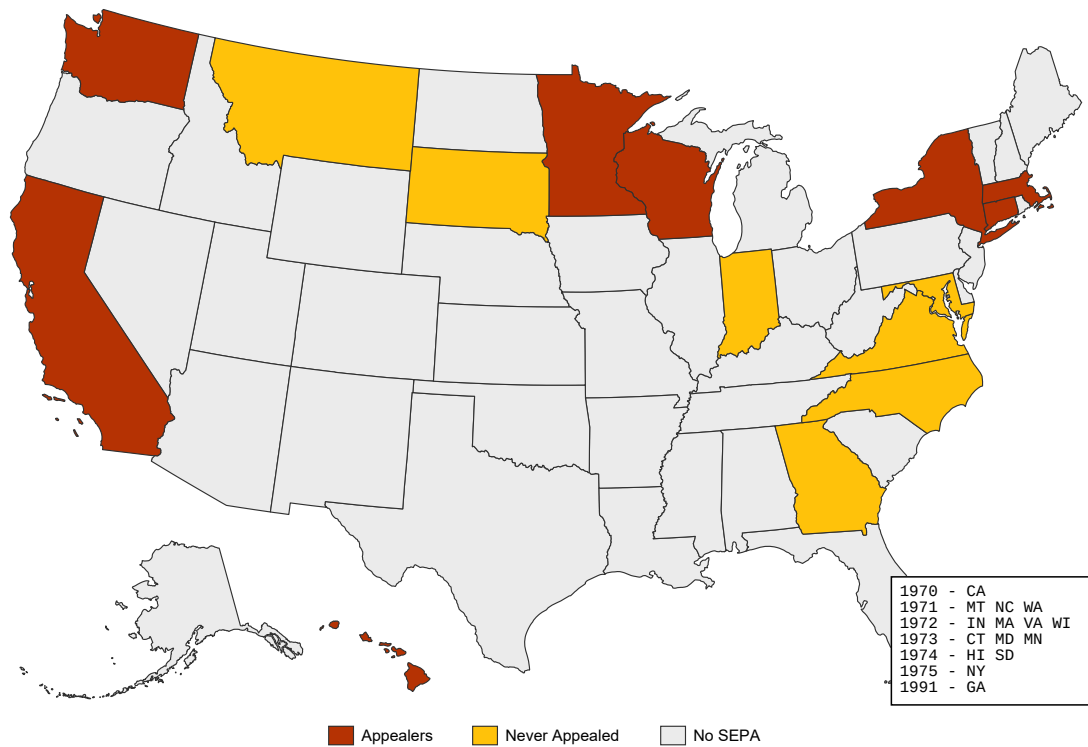
Notes. Executive orders that impose statewide EIR requirements are not treated as SEPAs because they cannot be used to begin a lawsuit. New Mexico passed a SEPA in 1971 but repealed it in 1974; it now has limited EIR requirements but is excluded from my historical analysis.

Table 1: States, SEPAs, and Early SEPA Cases

State	Authority	Cases	Earliest Residential Case	Year
California	Cal. Pub. Res. Code § 21000	1,874	Friends of Mammoth v. Mono County	1972
New York	N.Y. Envtl. Conserv. § 8-0101	928	Tuxedo Cons. & Taxpayers v. Tuxedo	1978
Washington	Wash. Rev. Code Ann. § 41.21C.030	193	Eastlake Council v. Roanoke Assoc.	1973
Minnesota	Minn. Stat. Ann. § 116D.04	148	Cedar-Riverside Env. Def. Fund v. Hills	1976
Massachusetts	Mass. Gen. Laws Ann. Ch. 30 § 61	107	Springfield Y Trust v. Exec. Dir. MHFA	1976
Connecticut	Conn. Gen. Stat. Ann. § 22a-1	76	Red Hill Co. v. Glastonbury Cons. Com'n	1989
Wisconsin	Wis. Stat. Ann. § 1.11	64	Wis. Env. Decade v. ILHR Dept.	1981
Hawai'i	Haw. Rev. Stat. § 343-5	63	McGlone v. Inaba	1981
Montana	Mont. Code Ann. § 75-1-201	29	N/A	N/A
North Carolina	N.C. Gen. Stat. § 113A-4	19	N/A	N/A
Maryland	Md. Nat. Res. § 1-301	11	N/A	N/A
South Dakota	S.D. Codified Laws § 34A-9-4	3	N/A	N/A
Georgia	Ga. Code Ann. § 12-16-4	2	N/A	N/A
Indiana	Ind. Code § 13-12-4-5	2	N/A	N/A
Virginia	Va. Code Ann. § 10.1-1188	1	N/A	N/A

Notes. Authority refers to the section of a SEPA cited the most times by cases reported in WestLaw as of 8 June 2026. Cases refers to the total number of reported decisions recorded in West's National Reporter System that cite the listed Authority. First Residential Case refers to the earliest case reported in the NRS that both 1) cited the given Authority and 2) involved plans to develop new residential units, where Year reports the year of the decision in that case and Success? denotes whether the final decision reversed issuance of a permit or otherwise further impeded the planned development.

Figure 3: States with SEPA's



Notes. Note box lists states whose SEPA's went into effect in a given year. Appealers are states in which a SEPA case involving a residential development project has been reported in WestLaw, as listed in Table 1. Never Appealed are SEPA states with no such cases.

Table 2: Historical Building Permit Summary Statistics by Block

	Total Permits	Permits/1K	SFH/1K	Mid/1K	Big/1K
<i>Appealers</i> (CA CT HI MA MN NY WA WI)					
Average, 1960–1993:	48,734.59	6.78	3.89	0.47	2.41
1960–1975	51,525.51	7.83	4.12	0.54	3.17
1976–1993	46,253.78	5.85	3.69	0.41	1.74
<i>Never Appealed</i> (GA IN MD MT NC SD VA)					
Average, 1960–1993:	27,333.30	6.20	4.10	0.30	1.80
1960–1975	24,815.12	6.23	3.72	0.26	2.25
1976–1993	29,571.69	6.18	4.44	0.34	1.41
<i>Some EIR</i> (21)					
Average, 1960–1993:	26,414.96	6.33	3.92	0.49	1.91
1960–1975	25,186.83	6.53	3.85	0.49	2.19
1976–1993	27,506.63	6.15	3.99	0.50	1.66
<i>No EIR</i> (13)					
Average, 1960–1993:	16,399.60	5.96	3.89	0.43	1.65
1960–1975	15,575.29	5.92	3.77	0.42	1.73
1976–1993	17,132.32	6.00	3.99	0.43	1.57

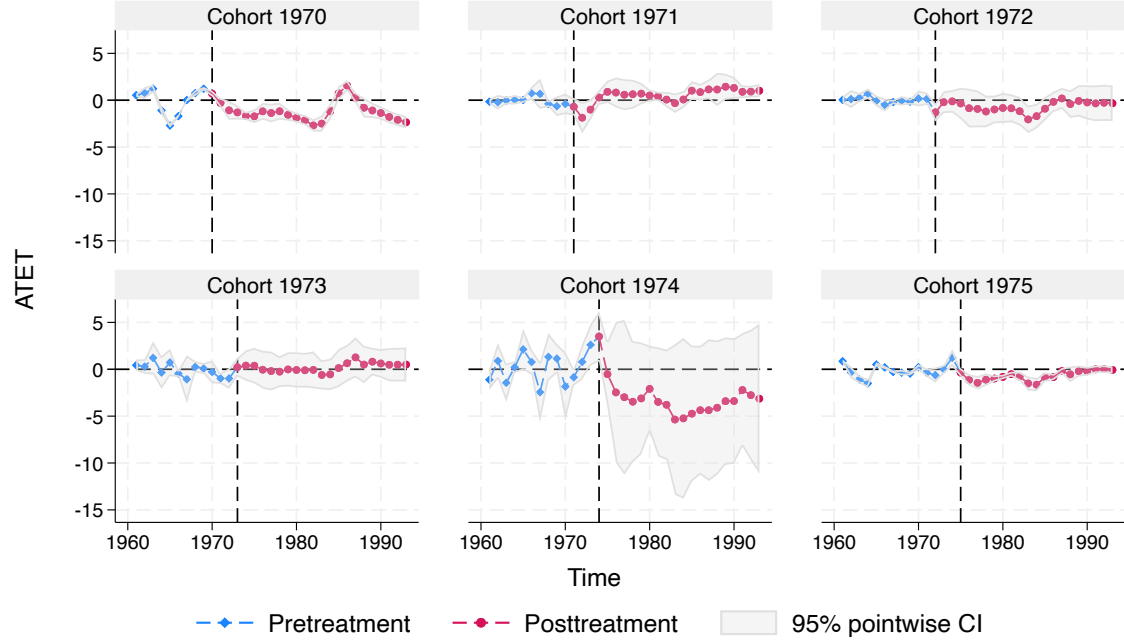
Notes. Base statistics are permits per year in a given state. Transformed permit outcomes are per capita based on population estimates in thousands (/1K = per 1,000) and averaged over states in a given block across the given time range. Blocks are as listed or following Figure 2. Mid = permitted units in buildings with 2-4 units. Big = permitted units in buildings with 5 or more dwelling units. Yearly population estimates come from Census Bureau estimates as reported by FRED; California estimates from FRED series CAPOP, Alabama from ALPOP, and so on.

Table 3: Main Spec SDID Results

	Units/1K (1)	SFH/1K (2)	Mid/1K (3)	Big/1K (4)
<i>Appealed</i> (CA CT HI MA MN NY WA WI)				
Donors: Some EIR	-1.338* (0.80)	-0.549* (0.33)	-0.125** (0.06)	-0.902** (0.43)
No EIR	-2.055* (1.09)	-0.480 (0.40)	-0.249** (0.11)	-1.379*** (0.54)
All non-SEPA	-1.528* (0.82)	-0.425 (0.29)	-0.144** (0.07)	-0.967** (0.44)
<i>Never Appealed</i> (GA IN MD MT NC SD VA)				
Donors: Some EIR	0.580 (0.51)	0.465* (0.27)	-0.003 (0.05)	-0.209 (0.58)
No EIR	0.047 (0.78)	0.553 (0.38)	-0.051 (0.07)	-0.773 (0.68)
All non-SEPA	0.396 (0.53)	0.575** (0.26)	-0.013 (0.06)	-0.358 (0.57)

Notes. Units/1K is total housing units permitted per 1,000 population in a given year. SFH refers to units built in single-family homes. Mid refers to permitted units in buildings with 2-4 units each. Big refers to permitted units in building with 5 or more units each. Estimates represent the average effect of having passed a SEPA on the given housing measure in the given SEPA block relative to a synthetic trend constructed from the donor pool listed on a given row. Bootstrapped standard errors (10,000 reps) in parentheses. 10%*, 5%***, 1%***.

Figure 4: Parallel Trends? via Callaway and Sant'Anna (2021)



Notes. Figure presents the baked-in Stata postestimation ATT graph for the doubly robust estimator described in Callaway and Sant'Anna (2021), comparing the cohort of SEPA states with a history of residential lawsuits to all non-SEPA states. Black dotted lines mark the first treated period, not any time before treatment.

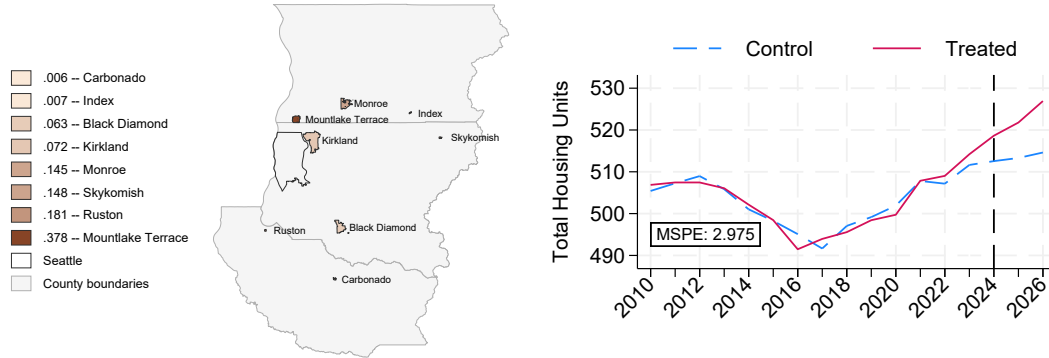
Table 4: Counterfactual Permit Counts

	CA	CT	HI	MA	MN	NY	WA	WI
Years, SEPA → 1993	24	21	20	22	21	19	23	22
Avg. Permits per Year	194,226	14,920	8,438	25,063	26,217	39,260	37,838	26,137
Avg. Counterfactual Permits	219,241	18,094	9,455	30,913	30,381	57,094	42,068	30,874
Permits / Counterfactual	.886	.825	.892	.811	.863	.688	.899	.847

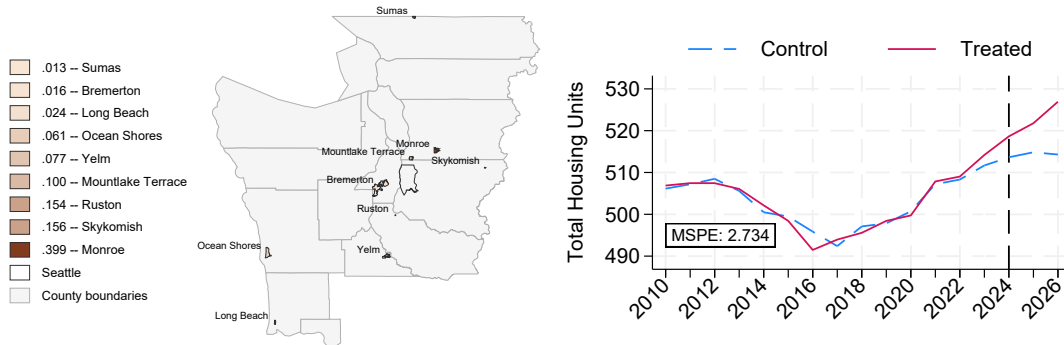
Notes. Table uses a -1 housing unit per 1K SEPA lawsuit risk coefficient estimate. Average Counterfactual Permits are calculated by taking observed total unit permit rates, subtracting the coefficient estimate of -1, multiplying the resulting rate by population in a given year, and taking the average across the years after a SEPA passed in a given state.

Figure 5: Synthetic Seattles

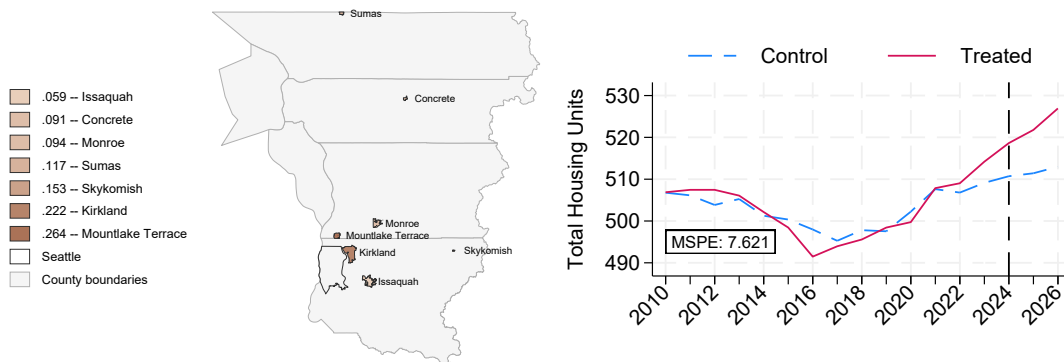
Seattle-Tacoma-Bellevue MSA



Coast or Sound Counties

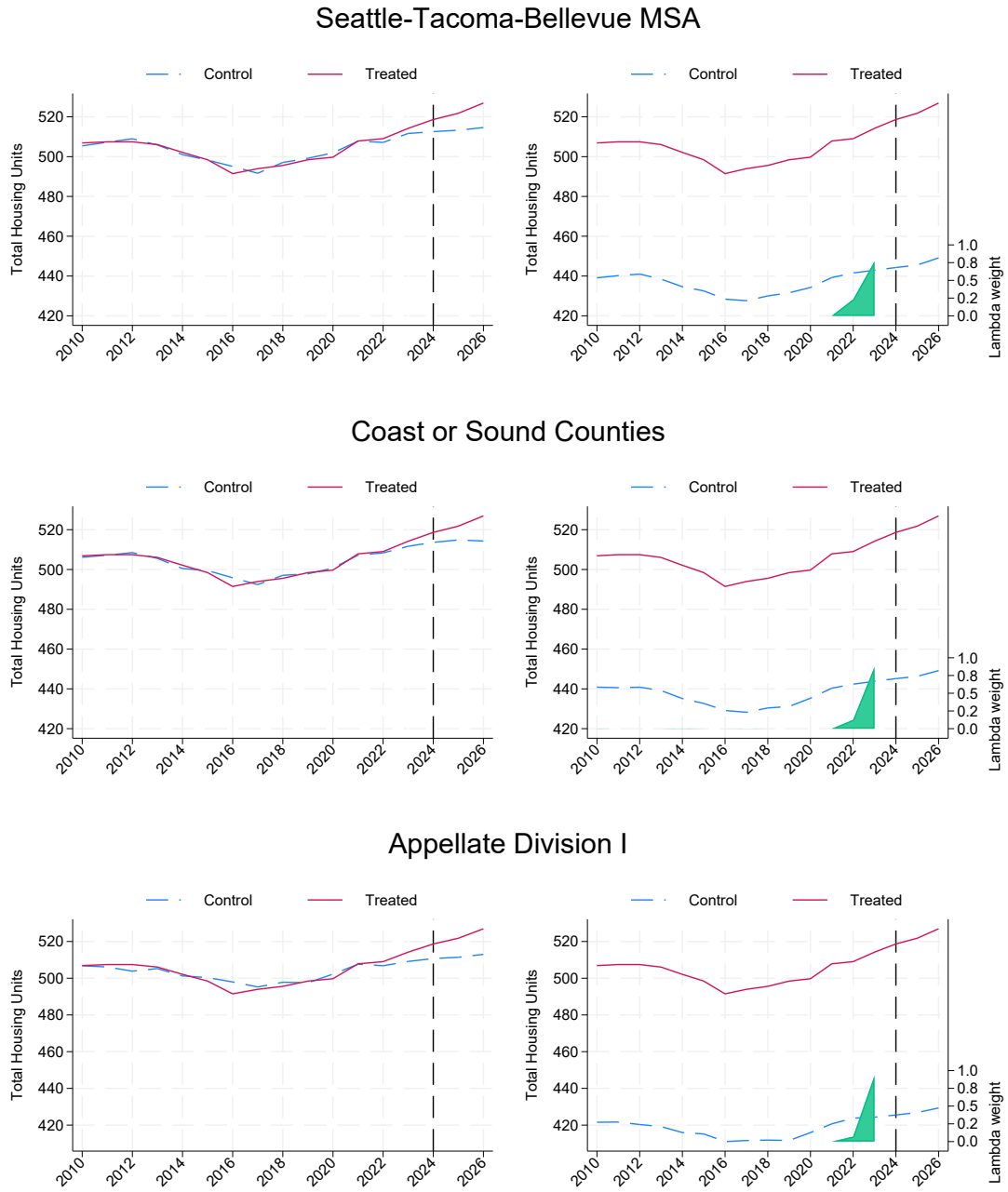


Appellate Division I



Notes. Seattle-Tacoma-Bellevue MSA includes King, Pierce, and Snohomish counties. Coast or Sound Counties border either the Pacific Ocean or the Puget Sound; the category includes Clallam, Grays Harbor, Island, Jefferson, King, Kitsap, Mason, Pacific, Pierce, San Juan, Skagit, Snohomish, Thurston, Wahkiakum, and Whatcom counties. Appellate Division I includes Island, King, San Juan, Skagit, Snohomish, and Whatcom counties. Weights in figures indicate the synthetic control weight assigned to listed cities for a given synthetic control construction; any unlisted city received zero weight. Black dotted lines mark the first treated period, not any time before treatment.

Figure 6: Synthetic Seattles Compared to SDID



Notes. Black dotted lines mark the first treated period, not any time before treatment.

Table 5: Washington State Housing Supply Summary Statistics by Comparison Group

	Population	Total Units	Units/1K
<u>Seattle</u>			
Average, 2010–2026:	707,476.00	358,536.18	506.24
2010–2019	656,827.70	328,728.90	500.79
2020–2023	755,278.75	383,535.50	507.70
2024–2026	812,566.67	424,561.33	522.45
<u>Seattle-Tacoma-Bellevue MSA</u>			
Average, 2010–2026:	26,487.55	10,889.50	412.42
2010–2019	25,102.95	10,381.53	414.68
2020–2023	28,043.95	11,378.32	407.68
2024–2026	29,027.65	11,930.95	411.24
<u>Coast or Sound Counties</u>			
Average, 2010–2026:	19,893.85	8,356.89	438.50
2010–2019	18,838.22	7,958.53	441.78
2020–2023	21,055.88	8,733.39	432.10
2024–2026	21,863.21	9,182.75	436.13
<u>Appellate Division I</u>			
Average, 2010–2026:	23,807.53	9,813.28	425.67
2010–2019	22,512.15	9,330.83	427.61
2020–2023	25,243.19	10,279.10	421.42
2024–2026	26,211.24	10,800.38	424.86

Notes. Base statistics are population or housing units present on April 1 of a given year in a given city. Rows after the “Seattle” row exclude Seattle. Units/1K is per capita based on population estimates in thousands (/1K = per 1,000) and averaged over cities in a given set of counties across the given time range. Seattle-Tacoma-Bellevue MSA includes King, Pierce, and Snohomish counties. Coast or Sound Counties border either the Pacific Ocean or the Puget Sound, and the category includes Clallam, Grays Harbor, Island, Jefferson, King, Kitsap, Mason, Pacific, Pierce, San Juan, Skagit, Snohomish, Thurston, Wahkiakum, and Whatcom counties. Appellate Division I includes Island, King, San Juan, Skagit, Snohomish, and Whatcom counties.

Table 6: Seattle Review Holiday Results

	SC (1)	SDID (2)
<u>Seattle-Tacoma-Bellevue MSA</u>		
Holiday Effect, 2024-2026	8.957 (40.64)	5.748 (3.56)
<u>Coast or Sound Counties</u>		
Holiday Effect, 2024-2026	8.190 (13.36)	5.669 (4.49)
<u>Appellate Division I</u>		
Holiday Effect, 2024-2026	10.743 (39.79)	5.658* (3.29)

Notes. Coefficient estimates represent the average effect of the Seattle holiday on housing units available per 1,000 population. Panels represent donor pools as described in the notes of Table 5. Bootstrap standard errors (10,000 reps) in parentheses. 10%*, 5%** , 1%***.

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